Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

N divisions on the main scale of a vernier calliper coincide with $(N+1)$ divisions of the vernier scale. If each division of main scale is ' a ' units, then the least count of the instrument is

(A) a

(B) $\frac{a}{N}$

(C) $\frac{N}{N+1} \times a$

(D) $\frac{a}{N+1}$

Q.2

Young's modulus of a material has the same unit as

(A) pressure

(B) strain

(C) compressibility

(D) force

Q.3

The mass and volume of a body are found to be $5.00 \pm 0.05 \mathrm{~kg}$ and $1.00 \pm 0.05 \mathrm{~m}^3$ respectively. Then the maximum possible percentage error in its density is

(A) 6%

(B) 3%

(C) 10%

(D) 5%

Q.4

In an experiment four quantities $\mathrm{a}, \mathrm{b}, \mathrm{c}$ and d are measured with percentage error $1\%, 2\%, 3\%$ and 4% respectively. Quantity P is calculated as $\mathrm{P}=\frac{a^3 b^2}{c d} \%$ error in P is

(A) 10%

(B) 7%

(C) 4%

(D) 14%

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Q.5

The respective number of significant figures for the numbers $23.023, 0.0003$ and 2.1×10^{-3} are

(A) $5, 1, 2$

(B) $5, 1, 5$

(C) $5, 5, 2$

(D) $4, 4, 2$

Q.6

A student measured the length of a rod as $3.50\ \text{cm}$. Which instrument did he use to measure it?

(A) A meter scale

(B) A vernier calliper where the 10 divisions in vernier scale matches with 9 divisions in main scale and main scale has 10 divisions in $1\ \text{cm}$

(C) A screw gauge having 100 divisions in the circular scale and pitch as $1\ \text{mm}$

(D) A screw gauge having 50 divisions in the circular scale and pitch as $1\ \text{mm}$

Q.7

The unit of impulse is the same as that of

(A) energy

(B) power

(C) momentum

(D) velocity

Q.8

In a simple pendulum experiment, the maximum percentage error in the measurement of length is 2% and that in the observation of the time-period is 3%. Then what is the maximum percentage error in determination of the acceleration due to gravity g ?

(A) 5%

(B) 6%

(C) 8%

(D) 10%

Q.9

Two full turns of the circular scale of a screw gauge cover a distance of $1\ \text{mm}$ on its main scale. The total number of divisions on the circular scale is 50 . Further, it is found that the screw gauge has a zero error of $-0.03\ \text{mm}$. While measuring the diameter of a thin wire, a student notes the main scale reading of $3\ \text{mm}$ and the number of

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circular scale divisions in line with the main scale as 35 . Then what is the diameter (in mm) of the wire?

(A) 3.38mm

(B) 3.67mm

(C) 2.85mm

(D) 3.40mm

Q.10

The density of material in CGS system of units is $4\ \mathrm{g/cm^3}$. In a system of units in which unit of length is $10\ \mathrm{cm}$ and unit of mass is $100\ \mathrm{g}$, then what will be the value of density of material?

(A) 0.4

(B) 40

(C) 400

(D) 0.04

Q.11

The current voltage relation of a diode is given by I = $\left(\mathrm{e}^{1000\ \mathrm{V/T}}-1\right)\ \mathrm{mA}$, where the applied voltage V is in volts and the temperature T is in degree kelvin. If a student makes an error measuring $\pm 0.01\ \mathrm{V}$ while measuring the current of 5 mA at $300\ \mathrm{K}$, what will be the error in the value of current in mA?

(A) 0.1mA

(B) 0.4mA

(C) 0.2mA

(D) 0.02mA

Q.12

The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is $0.5\ \mathrm{mm}$ and there are 50 divisions on the circular scale. The reading on the main scale is $2.5\ \mathrm{mm}$ and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 2%, then what is the relative percentage error in the density?

(A) 0.9%

(B) 2.4%

(C) 3.1%

(D) 4.2%

Q.13

When two quantities are divided, the relative error in the result is given by

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(A) the product of the relative error in the individual quantities

(B) the quotient of the relative error in the individual quantities

(C) the difference of the relative error in the individual quantities

(D) the sum of the relative error in the individual quantities

Q.14

If $Z = A^3$, then $\frac{\Delta Z}{Z} =$

(A) $\frac{\Delta \mathrm{A}^3}{\mathrm{A}}$

(B) $\left(\frac{\Delta \mathrm{A}}{\mathrm{A}}\right)^3$

(C) $3\left(\frac{\Delta \mathrm{A}}{\mathrm{A}}\right)$

(D) $\left(\frac{\Delta \mathrm{A}}{\mathrm{A}}\right)^{1/3}$

Q.15

The electric field is given by $\vec{E} = \frac{A}{x^3}\hat{i} + By\hat{j} + Cz^2\hat{k}$. The SI units of A, B and C are respectively: [where x, y and z are in m]

(A) $\frac{N-m^3}{C}, V/m^2,\ \mathrm{N/m^2 - C}$

(B) $\mathrm{V - m^2},\ \mathrm{V/m, N/m^2 - C}$

(C) $\mathrm{V/m^2},\ \mathrm{V/m, N - C/m^2}$

(D) $\mathrm{V/m, N - m^3/C, N - C/m}$

Q.16

If $\mathrm{L} = 2.331\ \mathrm{cm},\ \mathrm{B} = 2.1\ \mathrm{cm}$, then $\mathrm{L} + \mathrm{B} =$

(A) $4.4\ \mathrm{cm}$

(B) $4\ \mathrm{cm}$

(C) $4.43\ \mathrm{cm}$

(D) $4.431\ \mathrm{cm}$

Q.17

The speed of light in vacuum, c, depends on two fundamental constants, the permeability of free space, μ_0 and the permittivity of free space, ε_0. The speed of light is given by $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$. The units of ε_0 are $\mathrm{N^{-1}C^2\ m^{-2}}$. The units for μ_0 are

(A) $\mathrm{kg^{-1}\ m^{-1}C^2}$

(B) $\mathrm{kgmC^{-2}}$

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(C) $\mathrm{kgms^{-4}C^{-2}}$

(D) $\mathrm{kg^{-1}\ s^{-3}C^{-2}}$

Q.18

Number of significant figures in expression $\frac{4.327\ \mathrm{g}}{2.51\ \mathrm{cm^3}}$ is

(A) 2

(B) 4

(C) 3

(D) 5

Q.19

Error in the measurement of radius of a sphere is 1%. Then error in the measurement of volume is

(A) 1%

(B) 5%

(C) 3%

(D) 8%

Q.20

Find out the unit and dimensions of the constants a and b in the van der Waal's equation $\left(p+\frac{a}{V^2}\right)(V-b)=RT$, where p is pressure, v is volume, R is gas constant, and T is temperature.

(A) $A:\mathrm{Nm^4}, B:m^3$

(B) $A:m^3, B:Nm^4$

(C) $A:N^2m^3, B:m^4$

(D) $A:\mathrm{Nm^3}, B:m^3$

Answers & Solutions

Q.1 Answer:

$\frac{a}{N+1}$

Solution:

No of divisions on main scale $=N$ **No of divisions on vernier scale** $=N+1$ **size of main scale division** $=a$ **Let size of vernier scale division be** b **then we have** $aN=b(N+1)\Rightarrow \mathrm{b}=\frac{aN}{N+1}$
Least count is $\mathrm{a}-\mathrm{b}=\mathrm{a}-\frac{aN}{N+1}$
$=a\left[\frac{N+1-N}{N+1}\right]=\frac{a}{N+1}$

Q.2 Answer:

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pressure

Solution:

$$Y=\frac{\text{Stress}}{\text{Strain}}=\frac{\text{Force / Area}}{\text{Dimensionless}} \Rightarrow Y=\textbf{Pressure}$$

Q.3 Answer:

6%

Solution:

Here, Mass of a body, $\mathrm{M}=5.00 \pm 0.05 \mathrm{~kg}$ **Volume of a body,** $\mathrm{V}=1.00 \pm 0.05 \mathrm{~m}^3$
Density, $\rho=\frac{\mathrm{M}}{\mathrm{V}}$
Relative error in density is, $\frac{\Delta \rho}{\rho}=\frac{\Delta M}{M}+\frac{\Delta V}{V}$
Percentage error in density is $\frac{\Delta \rho}{\rho} \times 100=\frac{\Delta \mathrm{M}}{\mathrm{M}} \times 100+\frac{\Delta \mathrm{V}}{\mathrm{V}} \times 100$
$=\left(\frac{0.05}{5} \times 100\right)+\left(\frac{0.05}{1} \times 100\right)=1\%+5\%=6\%$

Q.4 Answer:

14%

Solution:

$$\mathrm{P}=\frac{\mathrm{a}^3 \mathrm{~b}^2}{\mathrm{cd}}, \frac{\Delta \mathrm{P}}{\mathrm{P}} \times 100\%=3 \frac{\Delta \mathrm{a}}{\mathrm{a}} \times 100\%+2 \frac{\Delta \mathrm{b}}{\mathrm{b}} \times$$
$$100\%+\frac{\Delta \mathrm{c}}{\mathrm{c}} \times 100\%+\frac{\Delta \mathrm{d}}{\mathrm{d}} \times 100\%$$
$$=3 \times 1\%+2 \times 2\%+3\%+4\%=14\%$$

Q.5 Answer:

$5,1,2$

Solution:

Number of significant figures in $23.023=5$ **Number of significant figures in** $0.0003=1$ **Number of significant figures in** $2.1 \times 10^{-3}=2$

Q.6 Answer:

A vernier calliper where the 10 divisions in vernier scale matches with 9 divisions in main scale and main scale has 10 divisions in $1 \mathrm{~cm}$

Solution:

Measured length of $\mathrm{rod}=3.50 \mathrm{~cm}$ **For vernier scale with 1 Main Scale Division** $=1 \mathrm{~mm}$ **9 Main Scale Division** $=10$ **Vernier Scale Division, Least count** $=1 \mathrm{MSD}-1 \mathrm{VSD}=0.1 \mathrm{~mm}$

Q.7 Answer:

momentum

Solution:

Impulse = change in momentum

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Q.8 Answer:

8%

Solution:

As we know, time period of a simple pendulum

$T = 2\pi\sqrt{\frac{L}{g}} \Rightarrow g = \frac{4\pi^2 L}{T^2}$

The maximum percentage error in g

$$\frac{\Delta g}{g} \times 100 = \frac{\Delta L}{L} \times 100 + 2\left(\frac{\Delta T}{T} \times 100\right)$$
$$= 2\% + 2(3\%) = 8\%$$

Q.9 Answer:

3.38mm

Solution:

Least count of screw gauge $= \frac{0.5}{50}\ \text{mm} = 0.01\ \text{mm}$ **.·. Reading** $=$ [**Main scale reading** $+$ **circular scale reading** $\times \text{L.C}]$ $-$ (**zero error**)

$= [3 + 35 \times 0.01] - (-0.03) = 3.38\ \text{mm}$

Q.10 Answer:

40

Solution:

In CGS system, $d = 4\frac{\text{g}}{\text{cm}^3}$

The unit of mass is $100\ \text{g}$ **and unit of length is** $10\ \text{cm}$**, so density** $= \frac{4\left(\frac{100\ \text{g}}{100}\right)}{\left(\frac{10}{10}\ \text{cm}\right)^3}$

$= \frac{\left(\frac{4}{100}\right)}{\left(\frac{1}{10}\right)^3}\frac{(100\ \text{g})}{(10\ \text{cm})^3}$

$= \frac{4}{100} \times (10)^3 \cdot \frac{100\ \text{g}}{(10\ \text{cm})^3}$

$= 40$ **unit**

Q.11 Answer:

0.2mA

Solution:

The current voltage relation of diode is $I = \left(e^{1000V/T} - 1\right)\text{mA}$ **(given)**

When, $\text{I} = 5\ \text{mA}, e^{1000\text{V/T}} = 6\ \text{mA}$

Also, $dI = \left(e^{1000V/T}\right) \times \frac{1000}{T}$

(By exponential function) $= (6mA) \times \frac{1000}{300} \times (0.01) = 0.2\ \text{mA}$

Q.12 Answer:

3.1%

Solution:

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Diameter $\mathrm{D}=$ **M.S.R.** $+($ **C.S.R** $)\times$ **L.C.** $\mathrm{D}=2.5+20\times\frac{0.5}{50}$

$\mathrm{D}=2.70\ \mathrm{mm}$

The uncertainty in the measurement of diameter $\Delta\mathrm{D}=0.01\ \mathrm{mm}$

We know that

$$\rho=\frac{\text{Mass}}{\text{Volume}}=\frac{M}{V}=\frac{M}{\frac{4}{3}\pi\left(\frac{D}{2}\right)^3}$$

$$\therefore\ \frac{\Delta\rho}{\rho}\times100=\frac{\Delta M}{M}\times100+3\frac{\Delta D}{\Delta}\times100$$

$$=2+3\times\frac{0.01}{2.70}\times100=3.1\%$$

Q.13 Answer:

the sum of the relative error in the individual quantities

Solution:

If $z=\frac{A}{B}$ **then** $\frac{\Delta z}{z}=\frac{\Delta A}{A}+\frac{\Delta B}{B}$

(d) The sum of the relative error in individval quantities.

Q.14 Answer:

$3\left(\frac{\Delta\mathrm{A}}{\mathrm{A}}\right)$

Solution:

as $\mathrm{z}=\mathrm{A}^3$**, so** $\frac{\Delta\mathrm{z}}{\mathrm{z}}=3\frac{\Delta\mathrm{A}}{\mathrm{A}}$

Q.15 Answer:

$\frac{N-m^3}{C}, V/m^2,\ \mathrm{N/m^2-C}$

Solution:

Unit of Electric field $(\mathrm{E})=(\mathrm{N/C})=(\mathrm{V/m})$

So, unit of $\mathrm{A}=\mathrm{E}\times\mathrm{x}^3=\frac{\mathrm{N-m^3}}{\mathrm{C}}=\left(\mathrm{V-m^2}\right)$

Unit of $\mathrm{B}=\mathrm{E/y}=\frac{N}{mc}=\left(\frac{V}{m^2}\right)$

Unit of $\mathrm{C}=\mathrm{E/z^2}=\frac{N}{m^2c}=\left(\frac{V}{m^3}\right)$

Q.16 Answer:

$4.4\ \mathrm{cm}$

Solution:

$\mathrm{L}+\mathrm{B}=2.331+2.1\cong4.4\ \mathrm{cm}$

Since minimum significant figure is 2**.**

Q.17 Answer:

$\mathrm{kgmC^{-2}}$

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Solution:

We have, $c=\frac{1}{\sqrt{\mu_0 \in_0}} \Rightarrow \mu_0=\frac{1}{\epsilon_0 c^2}$
units of $\mu_0=\frac{1}{\text{units of } \in_0 c^2}$
$=\frac{1}{N^{-1} C^2 m^{-2}(m s^{-1})^2}=N s^2 C^{-2}$
$=\left(\mathrm{kgms}^{-2}\right) s^2 C^{-2}=\mathrm{kgm} C^{-2}$

Q.18 Answer:

3

Solution:

Correct Answer: C

[c] In multiplication or division the final result should return as many significant figures as there are in the original number with the least significant figures.

Q.19 Answer:

3%

Solution:

$V=\frac{4}{3} \pi r^3$
$\frac{\Delta V}{V} \times 100=3\left(\frac{\Delta r}{r}\right) \times 100=3 \times 1\%=3\%$

Q.20 Answer:

$A: \mathrm{Nm}^4, B: m^3$

Solution:

We can add and subtract only like quantities.
$\Rightarrow$ Dimensions of $P=$ Dimensions of $\frac{a}{V^2}$ ...(i)
and dimensions of $v=$ Dimensions of b ...(ii)
From (i),
Dimensions of $a=$ Dimensions of $P \times$ Dimensions of V^2 $[a]=\left[M^1 L^{-1} T^{-2}\right] \times\left[L^3\right]^2=\left[M^1 L^5 T^{-2}\right]$
Unit of $a=$ Unit of $p \times$ Unit of $V^2=\frac{N}{\mathrm{m}^2} \times \mathrm{m}^6=\mathrm{Nm}^4$
From (ii), $[b]=[V]=\left[M^{\circ} L^3 T^0\right]$
So unit of $b=$ Unit of $V=\mathrm{m}^3$

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